

Summary Statistics by Scenario

Scenario	N Cats	$\cos(\theta)$ 68%	Angle 68% (°)	Mean Error (°)	Mean Clusters
best_case	133	0.917	23.5	24.2 ± 2.2	304.3
perfect_ct	132	0.957	16.9	21.0 ± 7.3	269.3
full_pipeline	132	0.816	35.3	53.7 ± 34.3	3585.8
weighted_linear	132	0.796	37.2	53.6 ± 33.1	3585.8
weighted_squared	132	0.817	35.2	51.9 ± 33.0	3585.8